

Question 1

Porins are transmembrane proteins found on the outer membrane of mitochondria. Their function is to transport substances across the membrane. **Fig. 1.1a** and **Fig. 1.1b** show the side view and top view of the structure of a single polypeptide of porin respectively.

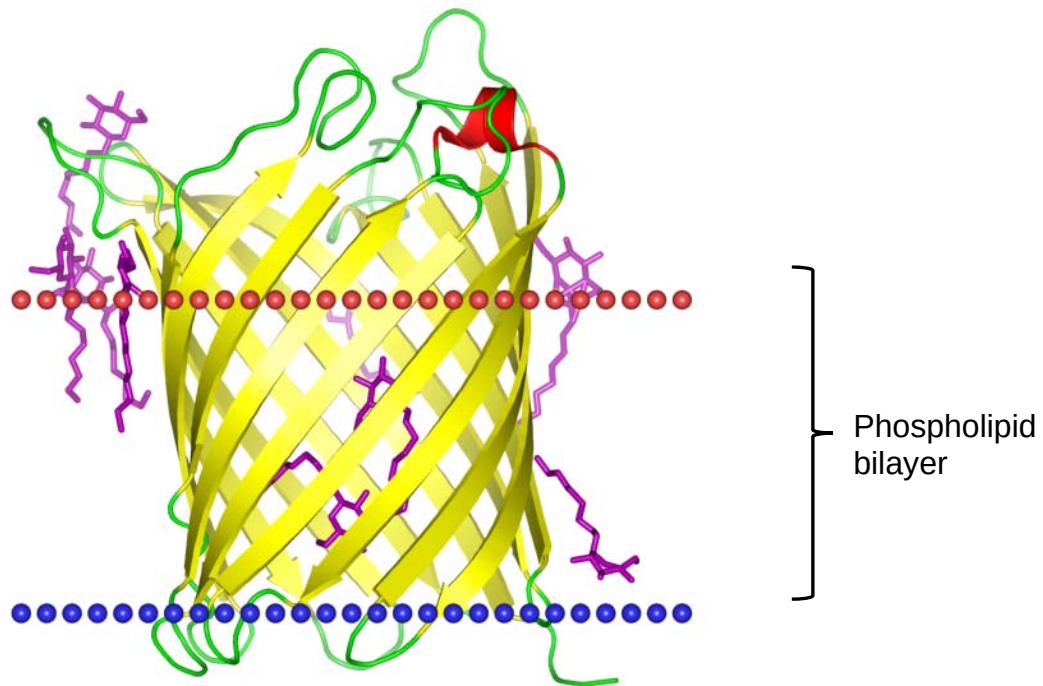


Fig. 1.1a



Fig. 1.1b

a) Name a covalent bond present within the protein porin. [1]

1. Peptide bond/disulfide bond

b) With reference to **Fig. 1.1a** and **Fig. 1.1b**,

(i) Name a non-covalent bond present within the protein porin. [1]

1. Hydrogen bond/ Ionic bond

(Reject hydrophobic interactions as not considered a bond)

(ii) Describe the structure of the protein porin. [2]

1. Contains **several anti-parallel β -pleated sheets**;

2. Folding of polypeptide chain to form specific 3D conformation/tertiary structure.

A researcher intends to study the transport of substances carried out by porins across the outer mitochondrial membrane. The researcher begins by isolating mitochondria from rat liver cells and placing them in 5 different solutions, each containing a different substance with a concentration of 10 arbitrary units as shown in **Table 1.2**.

Table 1.2

Solution	Content of Solution
1	10 AUs of Na ⁺ ions
2	10 AUs of Cl ⁻ ions
3	10 AUs of H ⁺ ions
4	10 AUs of protein insulin
5	10 AUs of acidic amino acid glutamate

The concentrations of the substances in the intermembrane space of the mitochondria were measured before the experiment and after 15 minutes of being placed in the solutions. The results obtained by the researcher are shown in **Table 1.3** on the following page (page 4).

Table 1.3

Solution	Content of Solution	Concentration of substances in mitochondrial intermembrane space before experiment	Concentration of substances in mitochondrial intermembrane space after 15 minutes
1	10 AUs of Na ⁺ ions	1 AU of Na ⁺ ions	10 AUs of Na ⁺ ions
2	10 AUs of Cl ⁻ ions	2 AUs of Cl ⁻ ions	10 AUs of Cl ⁻ ions
3	10 AUs of H ⁺ ions	2 AUs of H ⁺ ions	10 AUs of H ⁺ ions
4	10 AUs of protein insulin	0 AU of protein insulin	0 AU of protein insulin
5	10 AUs of acidic amino acid glutamate	0 AU of acidic amino acid glutamate	9 AUs of acidic amino acid glutamate

c) With reference to **Table 1.2** and **Table 1.3**,

- (i) Name the process by which the substances are transported through porins. [1]

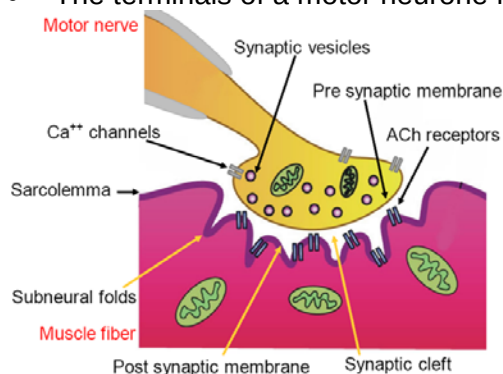
1. Facilitated diffusion

- (ii) Explain why insulin is unable to enter the mitochondrial intermembrane space. [2]

1. Insulin is a protein and is thus **too large** to enter via the porins.
2. Only charged substances and those of a small enough size/ quote examples from table (ions) can enter through the porins

d) Porins were artificially inserted into the membranes of a motor neurone. Predict and explain the effect of this modification on the function of the neurone. [4]

- The terminals of a motor neurone form synapses with the skeletal muscle fibres



1. Neurone will **not be able to transmit nervous implulses/no action potential** formed;
2. Effector/muscle/gland will not be stimulated;
3. **Resting membrane potential** is not maintained;
4. Due to **excess sodium ions influx into** the neurone down concentration gradient by facilitated diffusion

[Total: 11 marks]

Question 2

Scientists often employ the method of fluorescence imaging to study the process of nuclear division in cells. This method involves the use of fluorescence tagged antibodies to label specific structures within the cell, so that their movements and behaviours can be followed over the course of nuclear division.

A group of SR scientists decided to employ the aforementioned method to study the changes in centromere number in an animal cell undergoing meiosis. Initial testing showed that each centromere could be visualised as a single dot and therefore the total number of dots can be taken to correspond to the total number of centromeres within the cell. **Table 2.1** shows the number of dots and hence centromeres observed per cell, at various stages of meiosis.

Table 2.1

Stage of Meiosis	Prophase I	Prophase II	Anaphase II	Telophase II
Number of dots observed per cell	16			16

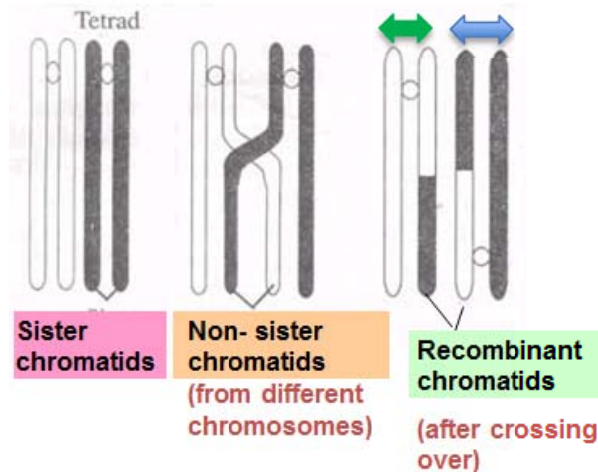
(a) With reference to **Table 2.1**,

- (i) Predict the number of dots (i.e. centromeres) observed in prophase II and anaphase II respectively. [2]

1. Prophase II: 8;
2. Anaphase II: 16

(ii) Explain your answer to part (a) (i). [2]

1. Prophase II: Homologous chromosomes are separated equally into 2 daughter cells/ haploid set of chromosomes is formed at the end of meiosis I/ following cytokinesis;
2. Anaphase II: Centromeres divide in anaphase II, thus doubling number of centromeres.

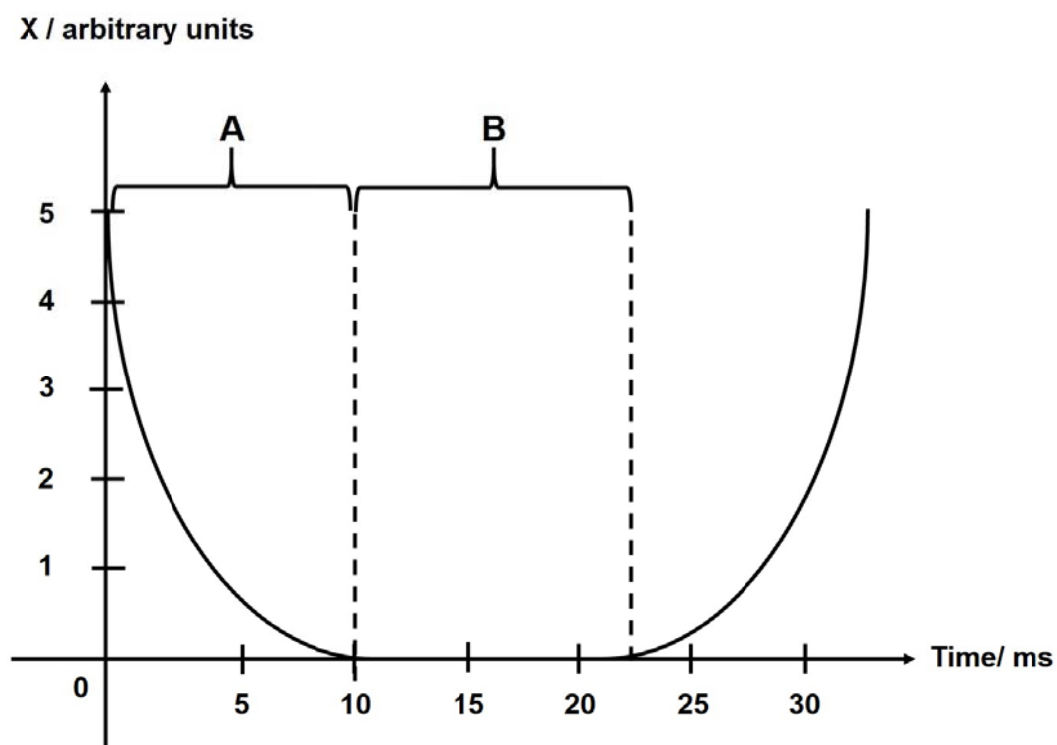


When the same experiment was conducted while the cell was in interphase, prior to meiosis I, no distinct dots could be observed in the nucleus. Instead, the entire nucleus emitted fluorescence.

(b) Suggest a reason for this observation. [1]

1. Chromatin had not condensed into discrete chromosomes with centromeres/ OWTTE.

Apart from using fluorescence imaging to determine the changes in centromere numbers, this technique was also employed by the SR scientists to study the movement of chromosomes during meiosis in the animal cell. More specifically, the changes in distance between 2 corresponding regions of non-sister chromatids of a pair of homologous chromosomes (**X**) were measured over the course of the first meiotic division. The results are shown in **Fig 2.1** below.



6
Fig 2.1

With reference to **Fig 2.1**, explain the changes in **X** for

(i) Region A. [2]

1. X decreases from 5 to 0 arbitrary units;
2. Homologous chromosomes pair up via synapsis to form bivalents

(ii) Region B. [2]

1. X remains constant at 0 arbitrary units;
2. Chiasma formation and crossing over occurs between non-sister chromatids of homologous chromosome

Driven by the success of the above experiment, one of the SR scientists decided to conduct the same experiment for the second meiotic division of the animal cell. His investigations however did not produce any results.

(c) State a reason for the lack of results obtained. [1]

1. Daughter cells are haploid/ only have 1 set of chromosomes/ OWTTE

1 mark each

[Total: 10 marks]

Question 3

Fig. 3.1 depicts a cell's responses to two kinds of DNA damage, a single stranded break and a double stranded break.

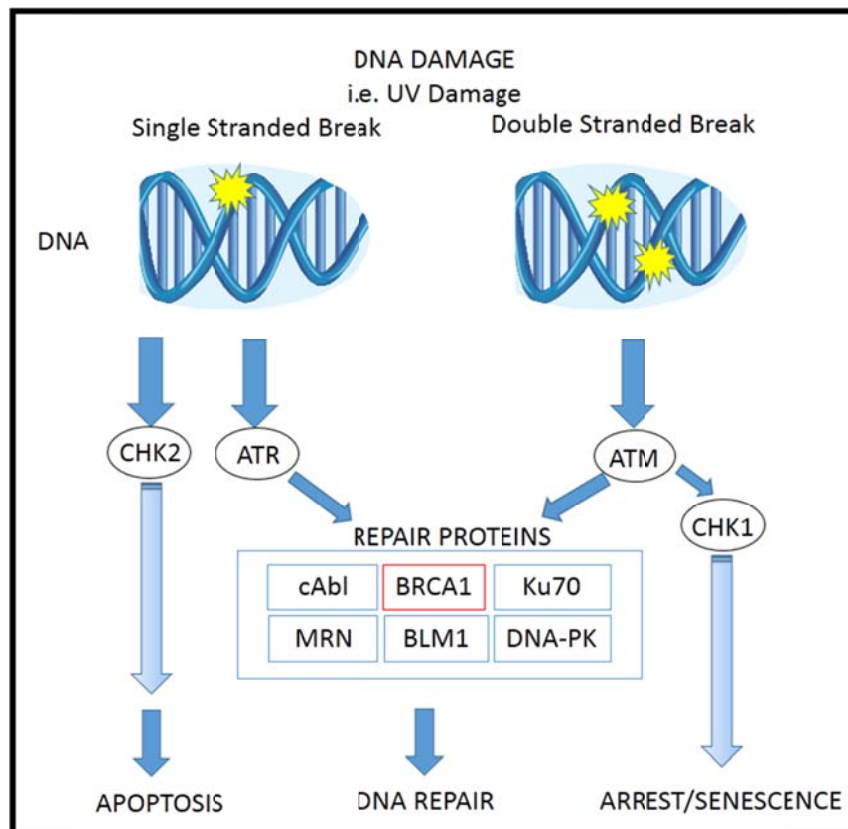


Fig. 3.1

(a) State the bond that would need to be broken to create both types of DNA damage. [1]

1. Phosphodiester Bond

(b) With reference to **Fig. 3.1**, describe the sequence of events that occurs in a cell that has a single strand break, eventually leading to DNA repair. [3]

- **Proteins such as ATR are activated;**
- **They facilitate the switching on expression of genes, thus BRCA 1/or any 1 named repair proteins are produced;**
- **Proteins produced when such genes are expressed help with DNA repair.**

(c) (i) Name the class of genes that produce the repair proteins. [1]

1. Tumour suppressor gene

(ii) Discuss the significance of this class of genes to cells. [2]

- These genes code for proteins that normally function to **prevent inappropriate cell cycle progression by inhibiting cell growth or division/ prevent uncontrolled division**;
- **Allows cells to prevent the progression of cancer via DNA repair, apoptosis & cell cycle arrest (required to list all as given in qn)**

(d) Describe one class of substances that can cause DNA damage and explain how they bring about the damage. [2]

- **Mutagens/Viruses/Ultra-violet/ Ionising Radiation (with specific example);**
- **Cause breakage in DNA/ distort double helix structure of DNA/ retroviruses integrate its genome and cause loss of function/gain of function mutations**

(e) Suggest a method by which cells will be able to repair a single strand break and explain why a double strand break may be harder to repair. [2]

- **Can be repaired using DNA ligase;**
- **Double strand break may lead to the DNA molecule splitting/ broken into two parts/fragments thus making it hard of DNA ligase to carry out ligation.**

[Total: 11 marks]

Question 4

Fig. 4.1 shows a plasmid commonly used to transfer genes to another bacterial cell via conjugation. This is a method for cloning genes into bacterial cells by using the plasmid of another bacterial cell as a vector. For conjugation to occur, the bacterial strain that passes the plasmid must have the genes necessary to build the conjugative apparatus (a pilus) between the bacterial cells.

This is normally carried on a large plasmid, called the F plasmid, that itself is transferred during conjugation, but due to its large size, it is not amenable to many such cloning strategies. In an attempt to still carry out cloning via conjugation in the laboratory without using the F plasmid, a small conjugative plasmid containing the origin sequence for conjugative transfer was created. However, this plasmid, as shown in **Fig. 4.1**, still does not contain the necessary genes to build the pilus.

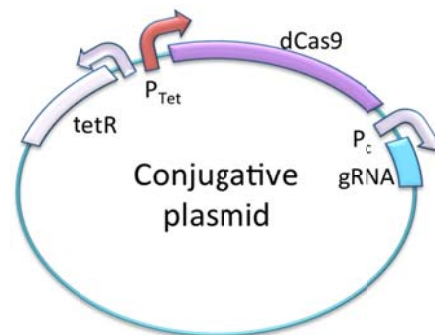


Fig. 4.1

(a) Describe the process of conjugation and elaborate on its significance. [4]

- **F+ donor cell synthesises sex pilus and make direct contact with F-recipient cell;**
- **Transfer of F plasmid from donor to recipient cell via formation of conjugation tube/cytoplasmic bridge (Not through sex pilus);**
- **Allows for genetic recombination to occur in bacteria/contributes to increased genetic variation in bacteria;**
- **Thus allows for bacteria to adapt to changing conditions**

(b) Describe how the F plasmid differs from the conjugative plasmid in **Fig. 4.1**. [3]

- **F Plasmid is larger than the smaller size conjugative plasmid;**
- **F plasmid carries the F factor DNA/ DNA that encode the synthesis of sex pili;**
- **F Plasmid does not have genes like dCas9, gRNA but these are found in the conjugative plasmid**

(c) As noted above, the conjugative plasmid does not have the required genes to produce the pilus. Assuming you only have this plasmid to use, suggest how you could overcome this issue and successfully transfer a gene on the conjugative plasmid via conjugation to another bacterial cell. [2]

- **Insert the gene of interest into the conjugative plasmid;**
- **But transform this plasmid into a host bacterial cell that carries the required genes for the pilus within its chromosomal DNA.**

(d) Suggest a use for the tetR gene on the conjugative plasmid. [2]

1. To select/ identify bacteria that has taken up the recombinant conjugative plasmid;
2. These bacteria will survive on medium containing tetracycline

[Total: 11 marks]

Question 5

Fig. 5.1 below shows the ears of maize (*Zea mays*). On the left, is variety Tom Thumb, the Black Mexican one on the right, and that of their F₁ hybrid in the centre.

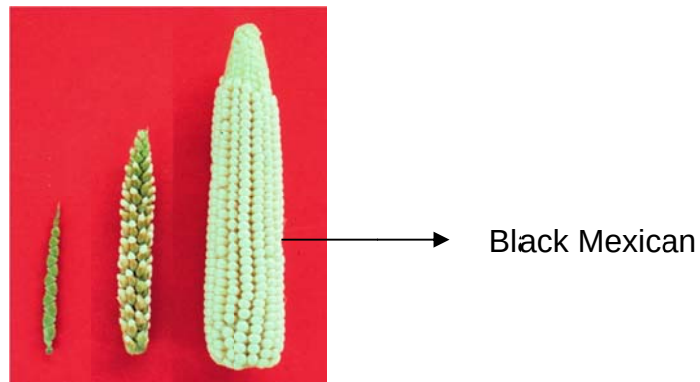


Fig. 5.1

(photo by John Doebley)

A cross was made between 2 pure breeding maize varieties, Tom Thumb and Black Mexican which differed markedly in ear length. The ear length of both parents, the F₁ and the F₂ generations, was measured in centimetre. This is given in the table below with the number of ears in each length category (e.g.14 of the F₁ plants produced ears 12 cm in length).

	Ear length (cm)																
	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Black Mexican parents									3	11	12	15	26	15	10	7	2
Tom Thumb parents	4	21	24	8													
F₁					1	12	12	14	17	9	4						
F₂			1	10	19	26	47	73	68	68	39	25	15	9	1		

Both parents showed variations in ear length. The variations in the F₁ are comparable to the average of the parental variations while the variations in the F₂ are greater than that found within the parental lines or the F₁.

(a) Suggest an explanation for this. [3]

1. **Variations in parental lines must be due to the effect of environment**
2. **as they are pure breeding, therefore must be homozygous at all loci / genetically identical**
3. **F₁ offspring should all be heterozygous at all loci/ genetically identical**
Variations must be due to the effect of environment. (award once)
4. **The greater variation is thus resulting not only from environment effects but also recombination leading to different combinations of alleles**
(name one example: independent assortment of homologous chromosomes/ crossing over & chiasmata formation/ random fertilization)

(b) Explain the genetic basis behind the wide range of ear lengths. [3]

1. **Additive effect of polygenes/ multiple genes;**
2. **Give rises to different combinations of genotypes that gives different intermediate phenotypes;**
3. **Each gene would have little overall effect;**
4. **Or give appropriate example: If ear length is controlled by e.g. 3 independent loci, (Tom parent will be homozygous –aabbcc alleles, and black Mexican – AABBCc alleles)**

F₁ fully heterozygous, AaBbCc shows intermediate ear length.

In F₂, the segregation & recombination of these 3 alleles pairs produce the greatest variations. E.g. AABBCc – longest.

(c) What is the term used to describe the range of phenotypes in the above example?
[1]

1. **Continuous variation**

[Total: 7 marks]

Question 6

In a college lab deep in opposition territory, an experiment was carried out using yeast cells (*Saccharomyces cerevisiae*) in anaerobic conditions. **Yeasts** are eukaryotic microorganisms classified in the kingdom Fungi, with 1,500 species currently described (estimated to be 1% of all fungal species). Yeasts are unicellular organisms, although some species with yeast forms may become multicellular. The experiment was set up as illustrated in the **Fig. 6.1**.

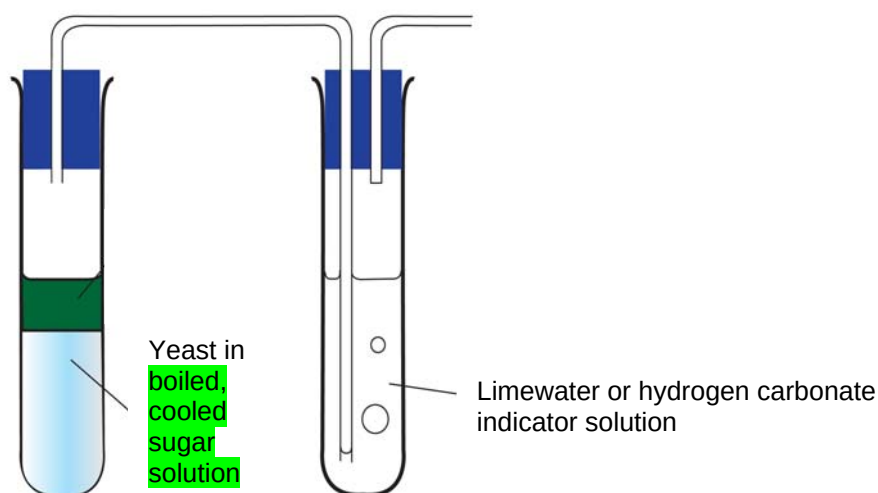


Fig. 6.1

(a) During the experiment, it was observed that the lime water turned cloudy. Explain how the yeast cells were able to bring about the cloudiness in the lime water. [3]

- **Breakdown of glucose to pyruvate in glycolysis**
- **Breakdown of pyruvate to ethanal releasing CO₂ via the enzyme pyruvate decarboxylase**
- **CO₂ reacts with limewater to bring about cloudiness.**

(b) A very conscientious student decided to do a follow-up experiment. He used the exact same experimental setup but replaced the yeast cells with cells from a mouse cell line. He found that the lime water did not turn cloudy. Comment on this observation. [2]

- **In animal cells, anaerobic respiration occurs via lactate fermentation instead of alcoholic fermentation which is the breakdown of pyruvate to lactate.**
- **In this process, CO₂ is not produced.**

In a separate experiment, the respiration rate of yeast was measured using a variety of substrates. The results are shown in **Fig. 6.2** below.

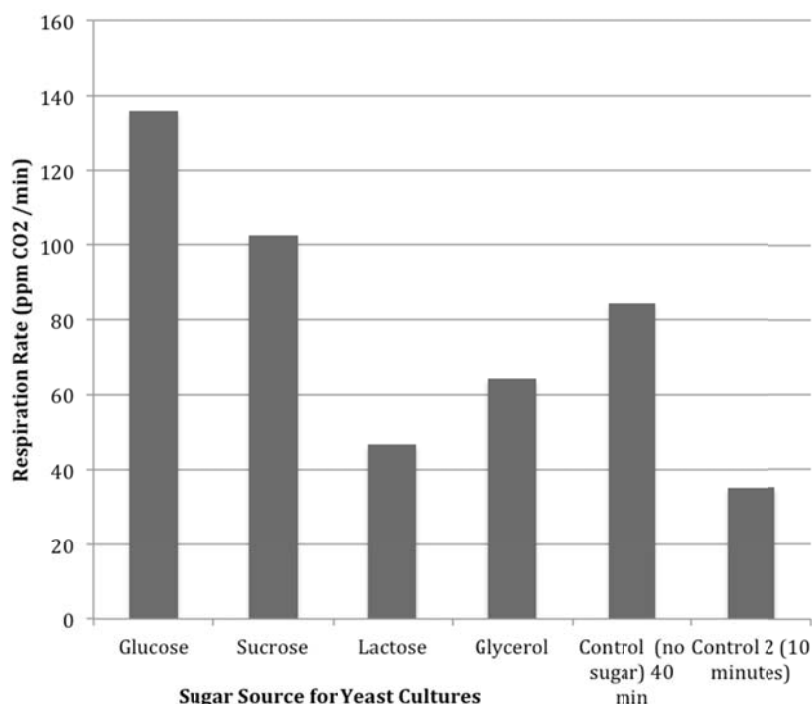


Fig. 6.2

(c) Explain why glucose appears to be a more effective respiratory substrate than sucrose. [3]

- **Glucose can be used directly in glycolysis of respiration**
- **Sucrose needs to be first broken down to give glucose and fructose.**
- **This additional step takes time hence the lower rate of respiration.**

(d) Suggest why using lactose as a substrate actually results in a lower respiration rate than control 1, which does not have any sugar provided. [2]

- **Yeast does not have the enzymes required to metabolise lactose.**
- **The respiration rate observed is due to glucose already present within the yeast / endogenous glucose.**

[Total: 10 marks]

Question 7

Fig. 7.1 shows the changes in beak size of medium ground finches on the Galapagos Islands over a period of about 8 years.

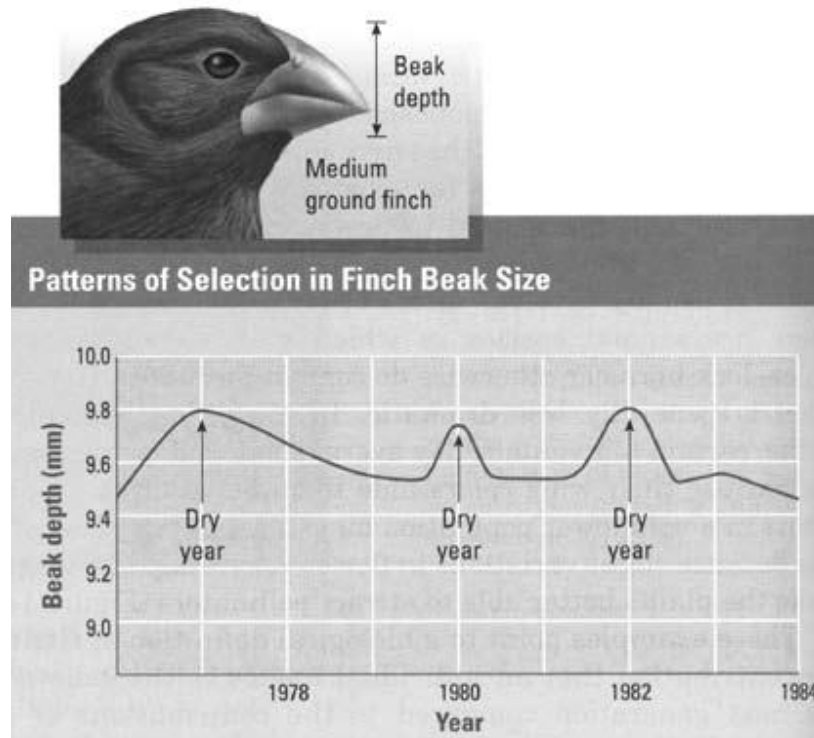


Fig. 7.1

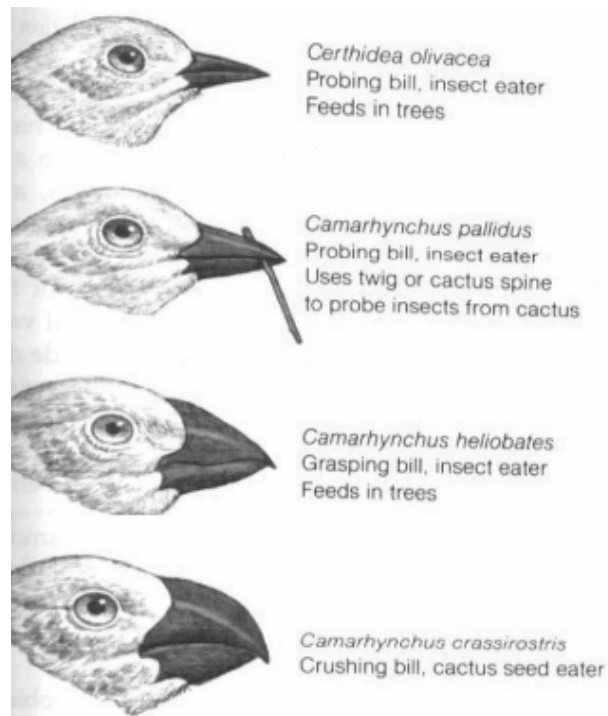
(a) (i) Describe the changes in beak size exhibited by the finches as reported in **Fig. 7.1**. [2]

- **Finches have a longer beak depth in dry years as compared to other years.**
- **Appropriate supporting data**

(ii) Suggest reasons for the changes described. [2]

- **Reference to appropriate reason (eg. food amount or food type) and sufficient elaboration linking to selection pressures.**

When scientists explored one of the islands, the following species of finches were observed.



(b) State the number of species of finches found on this island. [1]

- 4

It is believed that *Camarhynchus pallidus* is the original species to inhabit the island, having been forcibly blown in from the mainland during a particularly violent storm.

(c) Explain how this original ancestral species could have evolved into the current group of finch species found on the island. [4]

- **Variation present within the population**
- **Different selection pressure within different habitats**
- **New variation selected for and alleles for this variation passed to next generation.**
- **Genetic isolation/ no interbreeding between finches in the different habitats (adaptive radiation)**

Note: Traits are selected for but it is alleles that are passed down

Fig. 7.2 below shows the relationships between a number of finches including *Camarhynchus pallidus*. This relationship diagram was constructed using molecular data.

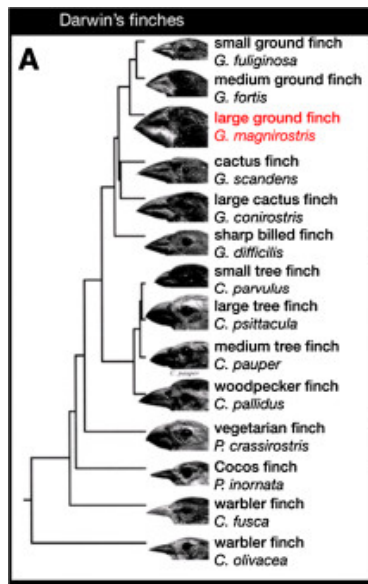


Fig. 7.2

(d) Name this relationship diagram and briefly explain how it could have been constructed. [3]

- **Cladogram/phylogenetic tree**
- **Identify a number of homologous features eg. DNA sequences, to compare between the species**
- **The more similar two species are/the more shared characteristics they have, the more closely they are related / share a more common recent ancestor. The more closely they are related, the closer their branching points should be to each other.**

(e) Describe the advantages of using molecular data in constructing this kind of diagram. [2]

- **No need to judge if potentially ambiguous structures are true homologies (e.g. the wings of a bat and the wings of a bird are not homologous structures).**
- **Large number of potential shared characteristics to compare (i.e. each nucleotide or amino acid position counts as one character).**
- **Can be used to compare very distantly related organisms which do not share any morphological or developmental homologies, e.g. plants and animals.**
- **A small amount of tissue sample is usually sufficient for analysis.**
- **DNA samples may be sometimes obtainable from well-preserved museum specimens, allowing the classification of rare or extinct species.**

Any 2

[Total: 14 marks]

Question 8

Multiple sclerosis (MS) involves an [immune-mediated process](#) in which an abnormal response of the body's immune system is directed against the central nervous system (CNS), which is made up of the brain, spinal cord and optic nerves. Within the CNS, the immune system attacks [myelin](#) — the fatty substance that surrounds and insulates the nerve fibers — as well as the nerve fibers themselves. **When any part of the myelin sheath or nerve fiber is damaged or destroyed, nerve impulses traveling to and from the brain and spinal cord are distorted or interrupted**, producing a wide variety of symptoms.

Adapted from The National Multiple Sclerosis Society Website

(a) Describe briefly the structure of a myelinated neurone. [2]

- **Presence of a myelin sheath covering axon/Dendron**
- **Presence of a cell body**
- **Presence of dendrites at one end and synaptic knobs at the other.**

Any 2

(b) Describe the normal function of a myelin sheath in nerve cells. [2]

- **As an electrical insulator**
- **to restrict action potential formation to nodes of Ranvier.**
- **Causes action potentials to 'jump' from one node of Ranvier to the next thus speeding up impulse transmission; saltatory conduction.**

Any 2

(c) Suggest how damage to the myelin sheath can lead to nerve impulses traveling to and from the brain and spinal cord becoming distorted or interrupted. [2]

- **Nerve impulses/action potentials may now cross between neighbouring neurons.**
- **Leads to interference of nervous impulses in these neighbouring neurons.**

[Total: 6 marks]

Question 9

9 (a) Explain the principles of homeostasis in terms of receptors, effectors and negative feedback. [7]

- 1 definition: maintenance of a constant internal environment;
 - 2 organism not adversely affected by external environment;
 - 3 optimal level known as reference point/set point;
 - 4 receptor/detector detects signal / change/deviation (from reference point);
 - 5 information relayed to control centre;
 - 6 (control centre is) where information compared with reference point
 - 7 control centre initiates response;
 - 8 effector returns system to optimal;
 - 9 by eliminating deviation/producing opposite effect;
 - 10 (linked to point 9) ref. to negative feedback;
- @ 1mk

9 (b) Compare and contrast competitive and non-competitive inhibitors and their effects on the rate of enzyme activity. [7]

- 1 Both serve to lower rate of enzyme activity
- 2 Both may be reversible @1 mk

Competitive	Non-competitive
3 Structural similarity to substrate	4 No structural similarity to substrate
5 Binds to active site	6 Binds to site other than active site
7 Competes with substrate for active site	8 Does not compete with substrate for active site
9 Prevents/lowers formation of ES complex	10 May or may not prevent formation of ES complex
11 Does not change conformation of enzyme upon binding	12 Changes conformation of enzyme upon binding
13 Increasing [S] concentration alleviates effect of inhibitor	14 Increasing [S] concentration does not alleviate effect of inhibitor
15 Max rate of reaction can be reached (with increased S concentration) / V_{max} unchanged	16 Max rate of reaction cannot be reached (with increased S concentration) / V_{max} reduced

@ ½ mk, award if attempt at valid comparison

Note:

Non-competitive inhibitors are not always allosteric inhibitors.

Non-competitive inhibitors are not always irreversible.

9 (c) Suggest similarities between the interaction of substrate with an enzyme and the interaction of ligand with a receptor. [6]

- 1 Both bind to specific regions of protein (idea of specific region/portion is imp);
- 2 Active site for enzyme; Binding site for receptor;
- 3 Both are complementary in shape to sites that they bind to;
- 4 Both will bind to protein via H bonds, ionic bonds, hydrophobic interactions (weak bonds);
- 5 Both interactions are not permanent;
- 6 Both can induce a conformational change in the protein when they bind;
- 7 Ref. to induced fit hypothesis for enzymes; Ref. to activation of receptor;

Question 10

(a) Using named examples, compare repressible and inducible systems of gene regulation. [10]

Differences:

	Inducible operon	Repressible operon
1) Named example	<u>Lac</u> operon	<u>Trp</u> operon
2 & 3) Differences in regulation	<u>Usually not expressed/ turned off;</u> but can be <u>stimulated/ induced</u> when <u>allolactose/inducer</u> molecule binds to <u>lac repressor</u>. Inactivates repressor	<u>Usually expressed/turned on;</u> but can be <u>inhibited/ repressed</u> when a <u>co-repressor/ tryptophan</u> binds to <u>trp repressor</u>. Activates repressor
4) Gene product of regulatory gene	Synthesizes <u>active repressor</u>	Synthesizes <u>inactive repressor</u>
5) Events when <u>lactose/ tryptophan is absent</u>	<u>Active</u> repressor binds operator, transcription <u>prevented/</u> operon <u>not expressed</u> .	<u>Inactive</u> repressor cannot bind operator, transcription <u>occurs/</u> operon <u>expressed</u> .
6 & 7) Events when <u>lactose/ tryptophan is present</u>	<u>Allolactose binds</u> repressor, <u>repressor becomes inactive</u>. <u>Inactive</u> repressor cannot bind operator, transcription <u>occurs/</u> operon <u>expressed</u>.	<u>Co-repressor (tryptophan) binds</u> repressor, <u>repressor becomes active</u>. <u>Active</u> repressor binds operator, transcription <u>prevented/</u> operon <u>not expressed</u>.

Similarity:

- Both systems are examples of the **negative control of gene regulation**. **Transcription of structural genes is prevented** when **active form of repressor binds operator**.

(b) Explain how prokaryotic genome structure differs from eukaryotes. [6]

Genome Level:

- Prokaryotic genome is **circular** while eukaryotic **genome is linear**.
- Prokaryotic genes are **grouped into operons** with one promoter controlling **expression of many genes** while eukaryotic genes **are not grouped into operons**, each **gene has its own promoter**.
- Prokaryotic DNA is **free floating in the cytoplasm** as a nucleoid. Eukaryotic genome exists in the **nucleus as distinct chromosomes**.

4. Prokaryotic DNA is made up mostly **of coding regions**, with **few repetitive/satellite DNA**. Only **3% of eukaryotic DNA is coding**. Most of it is **satellite DNA/repetitive DNA**.
5. Prokaryotic DNA is **not coiled** round **histone proteins**, only **supercoiled**. **Eukaryotic DNA is coiled around histone proteins and further folded and coiled around scaffold proteins**.
6. Prokaryotic DNA has **no telomeres**, while eukaryotic DNA **has telomeres at the end** to prevent DNA degradation due to replication.
7. Prokaryotic genome has no introns while eukaryotic **genome has introns**.

(c) Discuss the evolutionary significance of organizing genes into chromosomes.[4]

1. **Eukaryotes have genes organized to chromosomes** because they are **more complex, require better regulation of gene expression**.
2. **They also have larger genomes hence a need to replicate them fast provides evolutionary advantages**.
3. **Specialisation – Each chromosome** has genes coding for similar **traits/set of traits. Allows faster expression of a group of traits**.
4. **Presence of silencer genes and activator genes on different chromosomes** allows **better control of transcription**. (Activator proteins bind to enhancers while silencer proteins bind to silencers)
5. **Faster replication** as different chromosomes can be **replicated simultaneously**.
6. **Allows for crossing over and independent assortment to increase genetic variation**.